

10.7.69

EE25BTECH11020 - Darsh Pankaj Gajare

Question:

Let $x^2 + y^2 - 4x - 2y - 11 = 0$ be a circle. A pair of tangents from the point $(4, 5)$ with a pair of radii form a quadrilateral with area_____.

Solution:

$$\text{Conic: } \mathbf{x}^T V \mathbf{x} + 2\mathbf{u}^T \mathbf{x} + f = 0 \quad (1)$$

$$V = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad \mathbf{u} = \begin{pmatrix} -2 \\ -1 \end{pmatrix}, \quad f = -11 \quad (2)$$

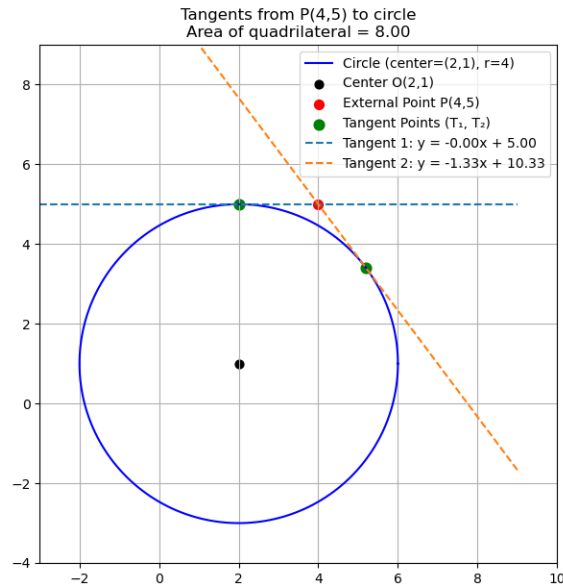
Equation of Tangent,

$$(\mathbf{V}\mathbf{q} + \mathbf{u})^T \mathbf{x} + \mathbf{u}^T \mathbf{q} + f = 0 \quad (3)$$

Point $\mathbf{a} = \begin{pmatrix} 4 \\ 5 \end{pmatrix}$ satisfies the given equation,

$$(\mathbf{V}\mathbf{q} + \mathbf{u})^T \mathbf{a} + \mathbf{u}^T \mathbf{q} + f = 0 \quad (4)$$

Plot using C libraries:



Plot using Python:

